CLASS TEST - I

(30)

(1) (a) $\ddot{x} - 6\dot{x} + 25x = 0$

Characteristic equatn :

$r^2 - 6r + 25 = 0$

$\Rightarrow r = \dfrac{6 \pm \sqrt{36 - \cancel{25}100}}{2}$

$= 3 \pm \cancel{\frac{8}{2}}\, 4i$

$\therefore \cancel{r_1 = \frac{9}{2}}$; $\cancel{r_2 = \frac{3}{2}}$ $r_1 = 3 + 4i$, $r_2 = 3 - 4i$

General solutn : $x(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}$

$\boxed{\cancel{x(t) = A_1 e^{\frac{9t}{2}} + A_2 e^{\frac{3t}{2}}}}$ ~~(Ans)~~

$= A_1 e^{(3+4i)t} + A_2 e^{(3-4i)t}$

$= A_1 e^{3t} \cdot e^{4it} + A_2 e^{3t} \cdot e^{-4it}$

$= \cancel{A_1 + A_2}\, e^{3t} [A_1 (\cos 4t + i \sin 4t) + A_2 (\cos 4t - i \sin 4t)]$

$= e^{3t} [(A_1 + A_2) \cos 4t + i\, \cancel{\sin} \sin 4t\, (A_1 - A_2)]$

$\therefore \boxed{x(t) = e^{3t} [B_1 \cos 4t + B_2 \sin 4t]}$ (Ans)

$B_1 = A_1 + A_2$; $B_2 = i(A_1 - A_2)$

(b) @ $x(0) = 1$.

$\Rightarrow \boxed{B_1 = 1}$ — (i)

@ $x(\pi) = 2 \Rightarrow e^{3\pi} [B_1] = 2$

$\Rightarrow \boxed{B_1 = 2e^{-3\pi}}$ — (ii)

(i) & (ii) contradict each other. Hence the ODE does not have a solutn for the given boundary conditions.

(c) @ $x(0) = 1 \Rightarrow \boxed{B_1 = 1}$.

$$x(t) = e^{3t}(\cos 4t + B_2 \sin 4t)$$

$$\Rightarrow \dot{x}(t) = e^{3t}(-4\sin 4t + 4B_2 \cos 4t + 3\cos 4t + 3B_2 \sin 4t)$$

$$= e^{3t}[(3B_2 - 4)\sin 4t + (4B_2 + 3)\cos 4t]$$

$\dot{x}(0) = -4$

$\Rightarrow 4B_2 + 3 = -4 \Rightarrow 4B_2 = -7$

$\Rightarrow \boxed{B_2 = -\frac{7}{4}}$.

$$\therefore \boxed{x(t) = e^{3t}\left(\cos 4t - \frac{7}{4}\sin 4t\right)} \quad \text{(Ans)}$$

(2) $$|M| = \begin{vmatrix} x & a & a & 1 \\ a & x & b & 1 \\ a & b & x & 1 \\ a & b & c & 1 \end{vmatrix}$$

$$= \begin{vmatrix} x-a & a-x & a-b & 0 \\ 0 & x-b & b-x & 0 \\ 0 & 0 & x-c & 0 \\ a & b & c & 1 \end{vmatrix}$$

$$= \begin{vmatrix} x-a & a-x & a-b \\ 0 & x-b & b-x \\ 0 & 0 & x-c \end{vmatrix}$$

$$= (x-a)(x-b)(x-c) = 0$$

$$\Rightarrow \boxed{x = a \text{ or } x = b \text{ or } x = c} \quad \text{(Ans)}$$

(3) (a) $D^2 = \frac{d^2}{dt^2}$

$$\therefore (D^2 + a^2)\sin \omega t = D^2 \sin \omega t + a^2 \sin \omega t$$

$$= D(\omega \cos \omega t) + a^2 \sin \omega t$$

$$= -\omega^2 \sin \omega t + a^2 \sin \omega t$$

$$\therefore \boxed{(D^2+a^2)\sin\omega t = (-\omega^2+a^2)\sin\omega t}$$

Hence, $\sin\omega t$ is an eigenfunction for the operator (D^2+a^2) with eigenvalue $(-\omega^2+a^2)$

Let $\cancel{f(D)}$ $f(D^2+a^2) = \dfrac{1}{D^2+a^2}$

From eigenvalue condition, if

$\cancel{D^2+a^2}$ $f(D)\sin\omega t = f(-\omega^2+a^2)\sin\omega t$

$$\Rightarrow \boxed{\frac{1}{D^2+a^2}\sin\omega t = \frac{1}{-\omega^2+a^2}\sin\omega t}$$ ~~(Ans)~~ (Proved)

(b) Let $x = \dfrac{1}{D^2+a^2}\sin at$

$\Rightarrow x(D^2+a^2) = \sin at.$

$$\Rightarrow \boxed{\ddot{x}+a^2x = \sin at} \quad \text{— (I)}$$

We try a particular solution of the form

$$\boxed{x_p = ct\cos(at)}$$

$\Rightarrow \dot{x}_p = c[\cos(at) - ta\sin(at)]$

$\Rightarrow \ddot{x}_p = c[-a\sin(at) - a\sin(at) - a^2t\cos(at)]$

$$\therefore \boxed{\ddot{x}_p + a^2x_p = -2ca\sin(at)} \quad \text{— (II)}$$

According to (I), we want the RHS of (II) $= \sin at$

$\therefore -2ca = 1 \Rightarrow \boxed{c = -\dfrac{1}{2a}}$

$\therefore \ddot{x}_p + a^2x_p = \sin at \Rightarrow \boxed{x_p = -\dfrac{t}{2a}\cos(at)}$

$$\therefore \boxed{\frac{1}{D^2+a^2}\sin at = \frac{-t}{2a}\cos at} \quad \text{(Ans)}$$

5 (a) $\lambda = 0$:

$\ddot{x} + \lambda x = 0 \Rightarrow \ddot{x} = 0$

$\Rightarrow \dot{x} = \cancel{at}\ a$

$\Rightarrow \boxed{x(t) = at + b}$

$x(0) = 0 \Rightarrow b = 0$

$x(L) = 0 \Rightarrow aL = 0 \Rightarrow a = 0$

$\therefore \boxed{x(t) = 0}$ (Proved)

* $\lambda < 0$:

$\ddot{x} + \lambda x = 0 \Rightarrow \ddot{x} = -\lambda x$.

Now, $-\lambda > 0$. Let $\boxed{-\lambda = k^2}$ — (i)

$\therefore \ddot{x} - k^2 x = 0$

General soln, $\boxed{x(t) = c_1 e^{kt} + c_2 e^{-kt}}$ — (ii)

$x(0) = 0 \Rightarrow c_1 + c_2 = 0 \Rightarrow \boxed{c_1 = -c_2}$

$x(L) = 0 \Rightarrow c_1 e^{-kL} + c_2 e^{kL} = 0$

$\Rightarrow c_1 e^{-kL} (\cancel{c_2}\ 1 - e^{2kL}) = 0$

$\because L > 0$ & $\lambda \neq 0$, hence $e^{2kL} \neq 1$. [from (i)]

$\therefore \boxed{c_1 = 0 \Rightarrow c_2 = 0}$

$\therefore \boxed{x(t) = c_1 e^{kt} + c_2 e^{-kt} = 0}$ [from (ii)]

(Proved)

(b) $\lambda > 0$:

$\ddot{x} + \lambda x = 0$

Let $\lambda = k^2$

$\therefore \ddot{x} + k^2 x = 0$

General soln : $\boxed{x(t) = A\cos(kt) + B\sin(kt)}$

$x(0) = 0 \Rightarrow \boxed{A = 0}$

$x(L) = 0 \Rightarrow B \sin(kL) = 0$

$\Rightarrow \sin(kL) = 0$ [$B \neq 0$ as that would mean $x(t) = 0$]

$\Rightarrow kL = n\pi \Rightarrow \boxed{k = \frac{n\pi}{L}}$

$\therefore \boxed{x(t) = B \sin\left(\frac{n\pi}{L} t\right)}$ (Ans)

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6 (a) $\ddot{x} + 3\dot{x} + 2x = 0$

$\Rightarrow$ Characteristic equat^n:

$r^2 + 3r + 2 = 0$

$\Rightarrow (r+2)(r+1) = 0.$

$\therefore r_1 = -2, \quad r_2 = -1$

$\therefore x(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}$

$\boxed{x(t) = A_1 e^{-2t} + A_2 e^{-t}}$

@ $t = 0$:

$x(0) = x_0 \Rightarrow A_1 + A_2 = x_0$ — (I)

$\dot{x}(0) = u \Rightarrow -2A_1 e^{-2t} - A_2 e^{-t} \Big|_{t=0} = u$

$\Rightarrow -2A_1 - A_2 = u$

$\Rightarrow 2A_1 + A_2 = -u$ — (II)

(II) − (I) $\Rightarrow$

$2A_1 + A_2 = -u$

$A_1 + A_2 = x_0$

(−) (−) (−)

$A_1 = -(u + x_0)$

$\therefore A_2 = x_0 - A_1$

$= x_0 + u + x_0 = 2x_0 + u$

$\therefore \boxed{x(t) = -(x_0 + u) e^{-2t} + (2x_0 + u) e^{-t}}$

@ $t \to \infty$: $e^{-2t} \to 0$ & $e^{-t} \to 0$, $\dot{x}(t) = 0$

$x(t) = 0$

Hence, the particle always comes to rest @ $x = 0$.

(b) e^{-2t} decays faster than e^{-t}. Hence, e^{-2t} goes faster to 0 as $t \to \infty$ than e^{-t}.

Hence, @ $t \to \infty$, only the coefficient of e^{-t} matters. The particle should approach from the left. This means that $x(t) < 0$.

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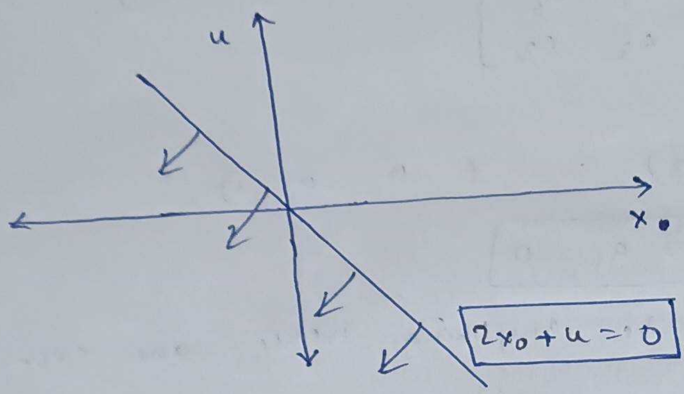
Coefficient of e^{-t} = $(2x_0 + u)$ in $x(t)$.

∴ $\boxed{2x_0 + u < 0}$ ⇒ $\cancel{\boxed{2x_0 < -u}}$



(c) @ $\cancel{t=0}$ ∴ $\boxed{x(t) = -2e^{-2t} + 3e^{-t}}$

∴ $\boxed{\dot{x}(t) = 4e^{-2t} - 3e^{-t}}$.

Starts moving to the left means $\dot{x}(t) = 0$, since it changes directⁿ.

∴ $\dot{x}(t) = 0 \Rightarrow 4e^{-2t} = 3e^{-t}$

$\Rightarrow 4 = 3e^{t}$

$\Rightarrow \boxed{t = \ln\left(\frac{4}{3}\right)}$ (Ans)

(7) (a) Let the matrix be $X = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

$\cancel{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}$ $\cancel{[\{3\} = X\{1\}]}$

$\Rightarrow \; = \begin{bmatrix} a_1 & \cancel{a_1} \\ \cancel{b_1} & \cancel{b_1} \\ c_1 & \cancel{c_1} \end{bmatrix}$

∴ ~~$a_3 = b_3 = c_3 = 1$~~. $a_1 = 0$, $b_1 = 0$, $c_1 = 1$

~~Other variables can have any value. Hence, one such matrix can be:~~

$$\cancel{X = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}}$$

$$\boxed{X = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}} \quad \text{(Ans)}$$

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$\therefore \langle 1 | X = \langle 3 |$

$$\Rightarrow (1 \ 0 \ 0) \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = (0 \ 0 \ 1).$$

$$\Rightarrow (a_1 \ a_2 \ a_3) = (0 \ 0 \ 1).$$

$$\Rightarrow \boxed{a_3 = 1}, \ \boxed{a_2 = a_1 = 0}$$

Other variables can be anything. Hence, one such matrix can be:

$$\boxed{X = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}} \quad \text{(Ans)}$$

(b) $$e^{i\sigma} = I + \frac{i\sigma}{1!} + \frac{(i\sigma)^2}{2!} + \frac{(i\sigma)^3}{3!} + \dots$$

$$= I + \frac{i\sigma}{1!} + -\frac{\sigma^2}{2!} - \frac{i\sigma^3}{3!} + \dots$$

$[\cancel{\sigma^2 = I \ ; \ \sigma^3 = \sigma^2\sigma = \sigma}]$

$$= I\left(1 - \frac{\sigma^2}{2!} + \dots\right) + i\left(\frac{\sigma}{1!} - \frac{\sigma^3}{3!} + \dots\right)$$

$$\boxed{e^{i\sigma} = \cos(\sigma) + i\sin(\sigma)} \quad \text{(Proved)}$$

(c) $$\cos(t\sigma) = I - \frac{(\sigma t)^2}{2!} + \frac{(\sigma t)^4}{4!} + \dots$$

$$= I - \frac{\sigma^2 t^2}{2!} + \frac{\sigma^4 t^4}{4!} + \dots$$

$$= I - \frac{t^2 I}{2!} + \frac{t^4 I}{4!} + \dots \qquad [\sigma^2 = I \Rightarrow \sigma^{2n} = I]$$

so, $\cos(t\sigma) = I\left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} + \dots\right)$

$= (\cos t)\, I$

$$\boxed{\cos(t\sigma) = \begin{bmatrix} \cos t & 0 \\ 0 & \cos t \end{bmatrix}}$$ (Ans)

(d) Let $\dot{x} = v$. The 2 differential equations are:

$\frac{dx}{dt} = v$ & $\frac{dv}{dt} = \ddot{x} = x$.

They can be combined & written in matrix form as:

$$\frac{d}{dt}\underbrace{\begin{bmatrix} x \\ v \end{bmatrix}}_{Y} = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{\sigma}\underbrace{\begin{bmatrix} x \\ v \end{bmatrix}}_{Y}$$

$\Rightarrow \frac{dY}{dt} = \sigma Y$

$\Rightarrow \boxed{Y(t) = e^{t\sigma} Y(0)}$ — (i)

$Y(0) = \begin{bmatrix} x(0) \\ v(0) \end{bmatrix} = \boxed{\begin{bmatrix} 0 \\ v_0 \end{bmatrix} = Y(0)}$ — (ii)

Now, $e^{t\sigma} = I + \frac{t\sigma}{1!} + \frac{(t\sigma)^2}{2!} + \frac{(t\sigma)^3}{3!} + \dots$

$= I + \frac{t\sigma}{1!} + \frac{t^2 I}{2!} + \frac{t^3\sigma}{3!} + \dots$

$[\ \sigma^2 = I \ ;\ \sigma^3 = \sigma^2\sigma = \sigma\ ]$

We have to diagonalise the matrix $t\sigma$.

$t\sigma = \begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix} = UDU^{-1}$

Eigenvalues:

$|t\sigma - \lambda I| = 0 \Rightarrow \begin{pmatrix} -\lambda & t \\ t & -\lambda \end{pmatrix} = 0$

$\Rightarrow \lambda^2 - t^2 = 0$

$\Rightarrow \boxed{\lambda = \pm t}$

$$\therefore \boxed{D = \begin{pmatrix} t & 0 \\ 0 & -t \end{pmatrix}} \quad \text{— (iii)}$$

Eigenvectors :

(i) $\lambda = t$:

Let the eigenvector be $A = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

$$\therefore (t\sigma - tI)\, A = 0$$

$$\Rightarrow \begin{bmatrix} -t & t \\ t & -t \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

$$\therefore \boxed{a_1 = a_2} \; ; \quad \boxed{A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}} \quad (\text{taking } a_1 = 1).$$

(ii) $\lambda = -t$:

Let the eigenvector be $B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

$$(t\sigma + tI)\, B = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = 0$$

$$\Rightarrow \boxed{b_1 = -b_2} \; ; \quad \boxed{B = \begin{pmatrix} 1 \\ -1 \end{pmatrix}}$$

$$\therefore U = \begin{pmatrix} A & B \end{pmatrix} = \boxed{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = U} \quad \text{— (iv)}$$

$$\therefore \boxed{U^{-1} = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}} \quad \text{— (v)}$$

For any function $\boxed{f(t\sigma) = U\, f(D)\, U^{-1}}$ — (vi)

$$\therefore e^{t\sigma} = U\, e^{D}\, U^{-1}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= -\begin{bmatrix} e^{t} & e^{-t} \\ e^{t} & -e^{-t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\therefore e^{t\sigma} = -\begin{bmatrix} e^t + e^{-t} & e^t - e^{-t} \\ e^t - e^{-t} & e^t + e^{-t} \end{bmatrix}.$$ (39)

$$\boxed{e^{t\sigma} = -2\begin{bmatrix} \cos t & i\sin t \\ i\sin t & \cos t \end{bmatrix}} \quad \text{— (vii)}$$

From (i), (ii) & (vii).

$$Y(t) = e^{t\sigma} Y(0)$$

$$= -2\begin{bmatrix} \cos t & i\sin t \\ i\sin t & \cos t \end{bmatrix}\begin{bmatrix} 0 \\ v_0 \end{bmatrix}$$

$$\Rightarrow Y(t) = -2\begin{bmatrix} i\sin t \; v_0 \\ \cos t \; v_0 \end{bmatrix} = \begin{bmatrix} x(t) \\ v(t) \end{bmatrix}$$

$$\therefore x(t) = -2i\sin t \; v_0$$

$$v(t) = -2\cos t \; v_0.$$

@ $x(1) = 1 \Rightarrow -2i\sin(1)\, v_0 = 1.$

$$\Rightarrow \boxed{v_0 = \frac{i}{2}\operatorname{cosec}(1)}$$

$$\therefore \boxed{\begin{aligned} x(t) &= \frac{\sin t}{\sin(1)} \\ v(t) &= -\frac{i\cos t}{\sin(1)} \end{aligned}} \quad \text{(Ans)}$$

(e) Matrix B :

$$|B - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 8 \\ 2 & -\lambda \end{vmatrix} = 0 \Rightarrow (\lambda - 3)\lambda - 16 = 0.$$

$$\Rightarrow \boxed{\lambda^2 - 3\lambda - 16 = 0} \quad \text{— (I)}$$

Matrix A :

$$|A - \sigma I| = 0 \Rightarrow \begin{vmatrix} a-\sigma & b \\ b & c-\sigma \end{vmatrix} = 0$$

$$\Rightarrow (a-\sigma)(c-\sigma) - b^2 = 0$$

$$\Rightarrow \boxed{\sigma^2 - (a+c)\sigma + (ac - b^2) = 0} \quad \text{— (II)}$$

Since A & B must have same eigenvalues, hence
(i) & (ii) should be similar. Hence, 40

$a+c = \boxed{3 \quad \& \quad ac - b^2 = -16}$

$\Rightarrow b^2 - ac = 16$

If we consider $c = 0$; $b = \pm 4$
$a = 3$

Taking $b = 4$, A becomes

$$\boxed{A = \begin{bmatrix} 3 & 4 \\ 4 & 0 \end{bmatrix}} \quad \text{(Ans)}$$

Hence, $\boxed{a = 3, \quad b = 4, \quad c = 0}$

(4) (a) Characteristic Equat^n:

$$r^2 + \beta r + \gamma = 0.$$

$$\therefore r = \frac{-\beta \pm \sqrt{\beta^2 - 4\gamma}}{2}$$

$$r = -\frac{\beta}{2} \pm \frac{\sqrt{\beta^2 - 4\gamma}}{2} \quad \begin{matrix} \rightarrow r_1 \rightarrow +ve \\ \rightarrow r_2 \rightarrow -ve \end{matrix}$$

General solut^n: $x(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}$

Case 1: $\cancel{\beta^2 - 4\gamma > 0}$ Solut^n diverges, if r_1 or $r_2 > 0$

$$-\frac{\beta}{2} \pm \frac{\sqrt{\beta^2 - 4\gamma}}{2} > 0.$$

$\Rightarrow \beta < \pm\sqrt{\beta^2 - 4\gamma}$ (a)

$\Rightarrow \beta^2 < \beta^2 - 4\gamma$ or $\beta^2 > \beta^2 - 4\gamma$ (b).

[Eg $4 > -5 \Rightarrow 4^2 < (-5)^2$

$3 > -2 \Rightarrow 3^2 > (-2)^2$]

If (a), $\gamma < 0$, & if (b), $\gamma > 0$

If